7 CHI-SQUARE TESTS

In the previous chapters, it was necessary to make certain assumptions about the population from which the samples were drawn. In many of the statistical tests, we had to assume that the samples came from normal populations. When this assumption cannot be justified, it is necessary to use procedures that do not require that these conditions be met. These procedures are generally referred to as non-parametric methods. In this chapter, we will discuss the χ^2 – test which belongs to this category. The χ^2 (pronounced as Chi-square) test is based on χ^2 – distribution which was first introduced by Karl Pearson in the year 1900.

The Chi-squareDistribution

For large sample size, the sampling (probability) distribution of χ^2 can be closely approximated by a continuous curve known as the Chi-square distribution. The probability function of χ^2 – distribution is given by:

$f(\chi^2) = c(\chi^2)^{\frac{v}{2}-1} e^{-\frac{\chi^2}{2}}$ where:

e = 2.71828, v = number of degrees of freedom and c = a constant depending only on v. The Chi-square distribution has only one parameter v, the number of degrees of freedom.

Important Properties of Chi-square Distribution

(1) χ^2 – distribution is a continuous probability distribution which has the value zero at its lower limit and extends to infinity in the positive direction. Negative values of χ^2 is not possible.

(2) The exact shape of the χ^2 – distribution depends upon the number of degrees of freedom v. For different values of v, we shall have

different shapes of the distribution. In general, when v is small, the shape of the curve is skewed to the right and as v gets larger, the distribution becomes more and more symmetrical and can be approximated by the normal distribution.

(3) The mean of the χ^2 – distribution is given by the degrees of freedom, i.e., $E(\chi^2) = v$ and variance is twice the degrees of freedom, i.e., $Var(\chi^2) = 2v$.

(4) As v gets larger, χ^2 – distribution approaches the normal distribution with mean = v and variance = $2v$.

(5) The sum of independent χ^2 variate is also a χ^2 variate. Therefore, if χ_1^2 is a variate with v_1 degrees of freedom and χ_2^2 is a χ^2 variate with v_2 degrees of freedom independent of χ_1^2, then their sum $\chi_1^2 + \chi_2^2$ is also a χ^2 variate with $v_1 + v_2$ degrees of freedom. This property is known as the additive property of χ^2.

Chi-square test

The χ^2 – test is one of the simplest and most widely used non-parametric tests in statistical work. It makes no assumptions about the population being sampled. The quantity χ^2 describes the magnitude of discrepancy between theory and observation, i.e., with the help of χ^2 – test we can know whether a given discrepancy between theory and observation can be attributed to chance or whether it results from the inadequacy of the theory to fit the observed facts. The formula for computing the Chi-square test is:

$\chi^2 = \sum \frac{(O-E)^2}{E}$, where:

O = observed frequency and E = expected or theoretical frequency.

The calculated value of χ^2 is compared with the table value of χ^2 for given degrees of freedom at specified level of significance. If the calculated value of χ^2 is greater than the table value, the difference between theory and observation is considered to be significant. On the other hand, if the calculated value of χ^2 is less than the table value, the difference between theory and observation is not considered to be significant.

The number of degrees of freedom is described as the number of observations that are free to vary after certain restrictions have been imposed on the data.

For a uniform distribution, we place one restriction on the expected distribution – the total of sample observations.

In a contingency table, the numbers of degrees of freedom are calculated as:

$(r-1)\times(c-1)$, where r = number of rows in the table and c = number of columns in the table.

Conditions for the Application of χ^2 – Test

The following five basic conditions must be met in order for Chi-square analysis be applied:

(1) The experimental data (sample observation) must be independent of each other.

(2) The sample data must be drawn at random from the target population.

(3) The data should be expressed in original units for convenience of comparison and not in percentage or ratio form.

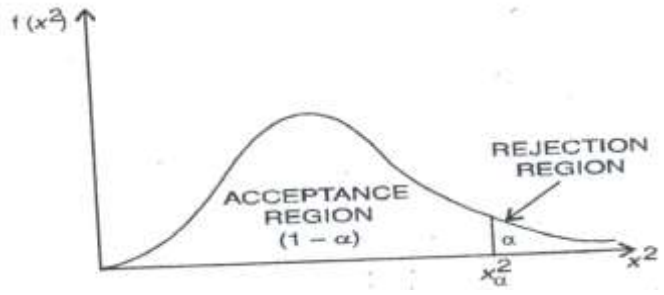
(4) The sample should contain at least 50 observations.

(5) There should not be less than five observations in any one cell (each data entry is known as a cell).

Use of Chi-square Table

To facilitate its many applications, the Chi-square distribution has been extensively

tabulated. The table of areas found in Chi-square tables gives values of χ^2 for variousprobabilities and various degrees of freedom. The value of is given in the columnheadings, the degrees of freedom,v are given in the rows and the body of the tablegives the χ^2 values. As depicted in the following figure, the value of χ^2 in the Chi-square table are given for various combinations of v and $1-\alpha$.



Grouping when Frequencies are small

If small theoretical frequencies occur (less than 10 and certainly not less than 5), it is generally possible to overcome the difficulty by grouping two or more classes together. In other words, one or more classes with theoretical frequencies less than 5 may be combined into a single category before calculating the difference between observed and

expected frequencies. The number of degrees of freedom would be determined with the number of observations after the regrouping. This would be clear from the following:

Observed frequencies	364	376	218	89	33	13	2	1
Expected frequencies	339	397	234	92	27	6	1	0

The last three classes should be combined together. After grouping, the position would be as follows:

Observed frequencies	364	376	218	89	33	16
Expected frequencies	339	397	234	92	27	7

The degrees of freedom would now be 6 – 1 = 5.

Some important applications of χ^2 – test are discussed in detail below:

(1) Tests of Hypothesis Concerning Variance

In testing hypothesis about the variance of a normally distributed population, the null hypothesis is $H_0 : \sigma^2 = \sigma_0^2$ against the alternative hypothesis $H_1 : \sigma^2 \neq \sigma_0^2$ where σ_0^2 is some specified value of the population variance.

To test the above hypothesis we compute $\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ where s^2 is computed from a random sample of size n.

Conclusion

If computed $\chi^2 > \chi^2_{\alpha/2}(n-1)$, we reject H_0, otherwise we accept H_0.

Example 1

Weights in kilograms of 10 shipments are given below:

38, 40, 45, 53, 47, 43, 55, 48, 52, 49.

Can we say that the variance of the distribution of weight of all shipments from which the above sample of 10 shipments was drawn is equal to 20 square kilogram?

Test at □ □ = 5% level of significance.

Solution

We want to test the hypothesis:

$H_0 : \sigma^2 = 20$ against the alternative hypothesis $H_1 : \sigma^2 \neq 20$.

To test the above hypothesis, we compute:

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}.$$

From our calculator, $s^2 = 31.111$ and $n = 10$. Hence

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2} = \frac{9 \times 31.1111}{20} = \frac{280}{20} = 14$$

The table value of χ^2 for 9 d.f. at 5% level of significance is 16.919.

Conclusion

Since computed $\chi^2 = 14 < \chi^2_{\alpha/2}(n-1) = 16.919$, we accept H_0. Hence, we conclude that the data are consistent with the hypothesis that the variance of the distribution of weights of all shipments in the population is 20 kilograms.

(2) Test of Independence

One of the most frequent uses of χ^2 is for testing the null hypothesis that two criteria of classification are independent. They are independent if the distribution of one criterion in no way depends on the distribution of the other criterion. If they are not independent, there is an association between the two criteria. In the test of independence, the population and sample are classified according to some attributes. The test will indicate

only, whether or not any dependency relationship exists between the attributes. It will not indicate the degree of association or the direction of dependency.

To conduct the test, a sample is drawn from the sample of interest and observed frequencies are cross-classified according to the two criteria. The cross-classification can be conveniently displayed by means of a table called a *contingency table*. Let us designate the two attributes as A and B where, attribute A is assumed to have *r* categories and attribute B is assumed to have *c* categories. Furthermore, assume the total number of observations in the problem is N. A representation of these observations is shown below in a table where O_{ij} represents the observation in the ith row and j^{th} column. Such a table in the matrix form is called a contingency table and is shown below.

	Attribute B							
Attribute A	B_1	B_2	B_3	…	B_j	…	B_c	Total
A_1	O_{11}	O_{12}	O_{13}	…	O_{1j}	…	O_{1c}	R_1
A_2	O_{21}	O_{22}	O_{23}	…	O_{2j}	…	O_{2c}	R_2
A_3	O_{31}	O_{32}	O_{33}	…	O_{3j}	…	O_{3c}	R_3
⋮	⋮	⋮	⋮		⋮		⋮	⋮
A_i	O_{i1}	O_{i2}	O_{i3}	…	O_{ij}	…	O_{ic}	R_i
⋮	⋮	⋮	⋮		⋮		⋮	⋮
A_r	O_{r1}	O_{r2}	O_{r3}	…	O_{rj}	…	O_{rc}	R_r
Total	C_1	C_2	C_3	…	C_j	…	C_c	N

In the table, R_i is the total of ith row and C_j is the total of jth column. The frequencies in these cells are termed as *cell frequencies* and the totals of

the frequencies in each of the rows (R_i) and columns (C_j) are termed as *marginal frequencies*.

Expected cell frequencies are computed according to the multiplicative rule of probability. If two events are independent, the probability of their joint occurrence is equal to the product of their individual probabilities. Applying this rule to a contingency table, it is equivalent to say that, if two criteria of classification are independent, a joint probability is equal to the product of the two corresponding marginal probabilities. Thus, the expected cell frequencies are given by the formula:

$$E_{ij} = \frac{R_i}{N} \times \frac{C_j}{N} \times N = \frac{R_i \times C_j}{N}$$

To conduct the test, same χ^2 is employed, i.e.,

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

will follow χ^2 - distribution with $v = (r-1) \times (c-1)$ degrees of freedom.

While applying the test, the null hypothesis is that the two attributes are independent.

If the calculated value of χ^2 is less than the table value at a specified level of significance, the null hypothesis holds true. If the calculated value of χ^2 is greater than the table value at a specified level of significance, the null hypothesis is rejected, i.e., the two attributes are associated.

Example 2

A sample of 200 persons with a particular disease was selected. Out of these, 100 were given a drug and the others were not given any drug. The results are as follows:

	Number of Persons

	Drug	No Drug	Total
Cured	65	55	120
Not cured	35	45	80
Total	100	100	200

Tests, whether the drug is effective or not.

Solution

Let us take the null hypothesis that the drug is not effective in curing the disease. Applying the χ^2- tests: the expected cell frequencies are computed as follows:

$$E_{11} = \frac{R_1 \times C_1}{N} = \frac{120 \times 100}{200} = 60; \qquad E_{12} = \frac{R_1 \times C_2}{N} = \frac{120 \times 100}{200} = 60$$

$$E_{21} = \frac{R_2 \times C_1}{N} = \frac{80 \times 100}{200} = 40; \qquad E_{22} = \frac{R_2 \times C_2}{N} = \frac{80 \times 100}{200} = 40$$

$$\chi^2 = \sum \frac{(O-E)^2}{E} = \frac{(65-60)^2}{60} + \frac{(55-60)^2}{60} + \frac{(35-40)^2}{40} + \frac{(45-40)^2}{40}$$
$$= 0.417 + 0.417 + 0.625 + 0.625 = 2.084$$

$$v = (r-1) \times (c-1) = (2-1) \times (2-1) = 1 \times 1 = 1$$

At □ = 5% level of significance; $\chi^2_{0.05}(1) = 3.84$

Conclusion

Since computed $\chi^2 = 2.084 < \chi^2_{0.05}(1) = 3.84$, we accept H_0. Hence, we conclude that the drug is not effective in curing the disease.